REFINED QUALITY ASSURANCE TESTING DOCUMENTATION (QATD)

Coordina AI

Autonomous AI Teammate & Workflow Orchestrator

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UM HACKATHON 2026

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Document Control

Field	Detail
System Under Test (SUT)	UM Hackathon 2026 - Coordina AI
Team Repo URL	https://github.com/jocelyngan2005/Coordina-AI.git
Project Board URL	https://sharing.clickup.com/90182652788/l/h/6-901817662468-1/4fe359d584ccd21
Live Deployment URL	https://coordina-ai.vercel.app/

Objective

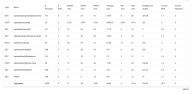
The primary objective is to evaluate the reliability and performance of Coordina AI, an autonomous AI Teammate & Workflow Orchestrator, in a complex environment. This includes testing its capability for document ingestion (PDF), multi-agent reasoning using the Gemini 3.1 Flash Lite Model, task planning, team coordination, risk detection, and submission readiness validation. The system must maintain performance thresholds under concurrent load and gracefully handle edge cases such as ambiguous inputs, inactive team members, and deadline failures.

Final Round (Execution, RCA & Hard Evidence)

1. **Test Execution & Evidence (Manual)**

Test Case ID	Test Type & Mapped Feature	Test Description	Expected Results	Actual Results	Status	Proof of Execution (Required for P0/ Critical tests)
TC-01	**Happy Case (Full Pipeline)**: Document → Analysis → Planning → Coordination → Risk	Verify the entire Coordina-AI workflow pipeline runs end-to-end successfully	The system should successfully ingest a PDF, generate structured goals, create a task list with milestones,	The backend stream-pipeline successfully orchestrates Analysis → Planning → Coordination → Risk. The	Pass	https://drive.google.com/drive/folders/1T0KPMfaGgN8vEOZdoU3-eDlcmYOzD-ki?usp=sharing

			assign initial roles, and redirect to the project workspace.	frontend receives SSE events and displays the progress in real-time.		
TC-02	**Negative (File Upload)**: Unsupported file type	Verify unsupported file type (.exe) falls back to raw decode and uploads successfully without crashing	Backend should detect unsupported MIME types (e.g., .exe) and fallback to utf-8 raw decoding (using errors='replace') to prevent server crashes during ingestion.	DocumentParser catches unsupported types and returns a raw decode string. The upload completes with 201 Created status, and the system continues to the next file without interruption.	Pass	https://drive.google.com/file/d/1_Rtj5nrcMgNXcWIXGpEkFkP4x-IcaD6N/view?usp=sharing
TC-03	**Negative (Workflow Ordering)**: Plan without prior analysis	Verify system blocks planning if no structured goals exist from prior analysis stage	Attempting to call the /plan (Planning) endpoint without first populating structured_goals (from the Analysis stage) should be blocked to prevent empty project states.	The WorkflowEngine explicitly checks for structured_goals. If missing, it throws a 422 Unprocessable Entity error: "Cannot plan: no structured goals found. Run analysis first."	Pass	https://drive.google.com/file/d/1P0foE_PPVKqg2cjUSvfOVTY14Cs3_TFb/view?usp=sharing
TC-04	**NFR (Performance)**: API	Verify read-only API	p95 < 800ms for all endpoints;	p95 < 50ms for all endpoints;	Pass	

	Response Time — Sequential	endpoints respond within 800ms p95 under sequential single-user load (100 requests each)	p50 < 400ms; 0% error rate	p50 < 50ms; all tests passed with no error		https://drive.google.com/file/d/1H4b0SdzRQZj5pvzsJ_Wid3PYjQVdl49y/view?usp=sharing
TC-05	**NFR (Load)**: Concurrent Users — Locust	Verify system handles 50 simulated concurrent users without degradation or errors	Error rate < 5%; p95 < 2500ms; zero HTTP 500 errors	Error rate = 0%; p95<2100ms; zero HTTP 500 errors	Pass	 https://drive.google.com/file/d/1PPPsUa_2lbgJ9dk2CGPOiD13HdoV7zSJ/view?usp=sharing

2. Automated Testing Pipeline

2.1. Unit Testing (Core Logic & Functions)

- **Test Runner Used:** Pytest
- **Targeted Components:**
  - inactivity_detector.scan()
  - deadline_recovery.generate_recovery_plan()
  - activity_tracker.get_member_last_activity()
  - agent._run()
  - workflow_engine.run_risk_check()
- **File(s) Covered:** https://github.com/jocelyngan2005/Coordina-AI/blob/main/backend/tests/unit/test_function_scenario.py

Unit Test ID	Function/Component Tested	Test Scenario	Expected Outcome	Actual Result	Status
UT-01	inactivity_detector.scan()	Tiered inactivity severity	Bob=critical, Alice=warm	Bob=critical, Alice=warm	**Pass**
UT-02	deadline_recovery.generate_recovery_plan()	Recovery plan structure	Tasks to cut: ['T4', 'T5']	Tasks to cut: ['T4', 'T5']	**Pass**

UT-03	activity_tracker.get_member_last_activity()	Most recent activity retrieval	Last activity: 2026-05-02T12:00:00Z	Last activity: 2026-05-02T12:00:00Z	**Pass**
UT-04	agent._run()	Critical risk urgency trigger	Urgency: immediate, Triggered: True	Urgency: immediate, Triggered: True	**Pass**
UT-05	workflow_engine.run_risk_check()	Risk check orchestration merge	Report has inactivity_report: True	Report has inactivity_report: True	**Pass**

Execution Proof:

```
tests\unit\test_function_scenario.py 2026-05-03 01:14:11 | WARNING | edge_cases.inactivity_detector - [InactivityDetector] Bob inactive for 5 d
s [critical]
2026-05-03 01:14:11 | WARNING | edge_cases.inactivity_detector - [InactivityDetector] Alice inactive for 2 days [warn]

============================================================
RUNNING TEST: UT-01
SCENARIO: Tiered inactivity severity
EXPECTED: Bob=critical, Alice=warn
ACTUAL:   Bob=critical, Alice=warn
STATUS:   SUCCESS
============================================================

2026-05-03 01:14:11 | INFO | edge_cases.deadline_recovery - [DeadlineRecovery] Generating recovery plan for project proj-123

============================================================
RUNNING TEST: UT-02
SCENARIO: Recovery plan structure
EXPECTED: Tasks to cut: ['T4', 'T5']
ACTUAL:   Tasks to cut: ['T4', 'T5']
STATUS:   SUCCESS
============================================================

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RUNNING TEST: UT-03
SCENARIO: Most recent activity retrieval
EXPECTED: Last activity: 2026-05-02T12:00:00Z
ACTUAL:   Last activity: 2026-05-02T12:00:00Z
STATUS:   SUCCESS
============================================================
```

```
============================================================
RUNNING TEST: UT-04
SCENARIO: Critical risk urgency trigger
EXPECTED: Urgency: immediate, Triggered: True
ACTUAL:   Urgency: immediate, Triggered: True
STATUS:   SUCCESS
============================================================

.2026-05-03 01:14:11 | WARNING | orchestrator.workflow_engine - [WorkflowEngine] 1 inactive members detected

============================================================
RUNNING TEST: UT-05
SCENARIO: Risk check orchestration merge
EXPECTED: Report has inactivity_report: True
ACTUAL:   Report has inactivity_report: True
STATUS:   SUCCESS
============================================================

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================================ 5 passed in 0.56s ================================
(.venv) PS C:\Users\Owner\Documents\competitions\umhack_2026\Coordina-AI\backend>
```

2.2. Integration & API Testing (System Modules)

- **Test Runner Used: Pytest**
- **Targeted Components:**
  - POST /api/workflow/[id]/analyse
  - POST /api/projects/ + POST /api/workflow/[id]/plan
  - POST /api/teams/
  - POST /api/workflow/[id]/risk-check -> GET /api/workflow/[id]/decisions
  - GET /api/analytics/project/[id]/overview
- **File(s) Covered:** https://github.com/jocelyngan2005/Coordina-AI/blob/main/backend/tests/integration/test_function_scenario.py

Test ID	Integration / Point Tested	Test Scenario	Expected Outcome	Actual Result	Status
IT-01	POST /api/workflow/[id]/analyse	Analysis flow integration	Status: success, Confidence: 0.95	Status: success, Confidence: 0.95	**Pass**
IT-02	POST /api/projects/ + POST /api/workflow/[id]/plan	Project and plan persistence	Tasks count: 1, Agent: planning	Tasks count: 1, Agent: planning	**Pass**
IT-03	POST /api/teams/	Project and plan persistence	Tasks count: 1, Agent: planning	Tasks count: 1, Agent: planning	**Pass**
IT-04	IT-04: POST /api/workflow/[id]/risk-check -> GET /api/workflow/[id]/decisions	Risk check decision logging	Log summary: Detected health: at_risk	Log summary: Detected health: at_risk	**Pass**
IT-05	GET /api/analytics/project/[id]/overview	Analytics aggregation integration	Health: critical, Readiness: 45	Health: critical, Readiness: 45	**Pass**

Execution Proof:

```
tests\integration\test_function_scenario.py
============================================================
RUNNING INTEGRATION TEST: IT-01
SCENARIO: Analysis flow integration
EXPECTED: Status: success, Confidence: 0.95
ACTUAL:   Status: success, Confidence: 0.95
STATUS:   SUCCESS
============================================================

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RUNNING INTEGRATION TEST: IT-02
SCENARIO: Project and plan persistence
EXPECTED: Tasks count: 1, Agent: planning
ACTUAL:   Tasks count: 1, Agent: planning
STATUS:   SUCCESS
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RUNNING INTEGRATION TEST: IT-03
SCENARIO: Team assignment integration
EXPECTED: Member ID: member-1, Status Code: 201
ACTUAL:   Member ID: member-1, Status Code: 201
STATUS:   SUCCESS
============================================================
```

```
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============================================================
RUNNING INTEGRATION TEST: IT-04
SCENARIO: Risk check decision logging
EXPECTED: Log summary: Detected health: at_risk
ACTUAL:   Log summary: Detected health: at_risk
STATUS:   SUCCESS
============================================================

RUNNING INTEGRATION TEST: IT-05
SCENARIO: Analytics aggregation integration
EXPECTED: Health: critical, Readiness: 45
ACTUAL:   Health: critical, Readiness: 45
STATUS:   SUCCESS
============================================================
```

3. Defect Log & Fix Traceability (Root Cause Analysis)

Bug ID	Bug Description	Steps to Reproduce	Console/Error Log Snippet	Root Cause Analysis (Why did it break?)	Fix Commit Hash / PR Link
DEF-01	Mismatch between frontend agent pipeline progress display and actual backend	1. Open a project workspace. 2. Update a task status. 3. Refresh the page. 4. Task reverts to previous status.	PUT /api/tasks/task-01 404 (Not Found)	The frontend was using temporary, agent-generated string IDs (e.g., task-01) for status updates instead of mapping them to the persistent database UUIDs.	5cb1676
DEF-02	Sidebar not displaying new projects	1. Create a new project from the dashboard. 2. Verify if the project appears in the sidebar navigation. 3. Observe it only appears after a hard refresh.	N/A (Stale state)	The ProjectsContext provider was not being triggered to re-fetch the project list (refreshProjects()) after successful project creation or deletion.	bd603d5
DEF-03	Analysis results not displaying on project workspace page	1. Navigate to a project with existing agent analysis. 2. Observe that the "Structured Goals" panel is empty or hidden.	undefined is not an object (evaluating 'state.goals.map')	Schema Mismatch: The analysis agent was returning results under structured_goals, while the frontend AnalysisPanel was initially looking for a goals key.	9658483

4. Known Issues & Deferred Technical Debt

Bug ID	Description & Impact (Bug or Technical Debt)	Reason for Deferral	GitHub Issue Link
DEF-04	UI Glitch: Blank page when switching from AdminProjectDetailPage to home to project team view. Causes temporary navigation stall.	Low Priority: Workaround exists by using the sidebar navigation; root cause is a state cleanup race condition in useEffect.	https://github.com/jocelyngan2005/Coordina-AI/issues/15
DEF-05	Static AI Activity Feed: The Admin Dashboard's "AI Agent Activity" feed currently uses hardcoded mock data (MOCK_ACTIVITY) instead of a real-time event stream.	Technical Debt: Backend event streaming was prioritized for the Project Workspace; Admin-side real-time monitoring is deferred to v1.2.	https://github.com/jocelyngan2005/Coordina-AI/issues/17
DEF-06	Read-Only Admin View: The Admin Project Detail page is strictly "Read Only" (as tagged in AdminProjectDetailPage.tsx: 581) and lacks override controls.	Strategic Deferral: Team lead autonomy is a core principle for the MVP; manual administrative overrides require a granular permission system.	https://github.com/jocelyngan2005/Coordina-AI/issues/18

5. Live Demo / UAT Risks

- **Risk 1: PDF-Only Dependency**
  The InstructionAnalysisAgent uses specific parsing logic optimized for PDFs. Uploading .docx, .txt, or image files will cause a parsing failure or return empty structured goals.
- **Risk 2: Cold Start / LLM Latency**
  Orchestrating 5 agents (Analysis → Planning → Coordination) involves multiple serial LLM calls. Total generation time can take 60–120 seconds; ensure the UI loading states are visible to the audience.
- **Risk 3: False Positive Inactivity Alerts**
  The RiskDetectionAgent polls Redis for activity signals. If a demo user is added to the team but doesn't "Check In" or "Update Task" immediately, the agent may flag them as a Critical Risk within minutes, which might look like a bug to a first-time viewer.
- **Risk 4: External API Quota Limits**

High-frequency interaction during UAT (e.g., 50+ users) can hit Google Gemini's rate limits. While we optimized connection pooling, external 429 errors will still surface as "Agent Busy" or "Risk Check Failed" messages.